SENTIENTIA LMS

Ninja Server Deployment & AWS RDS Installation Guidebook

Moodle 5.2 fresh install | Complete standalone package

Document	Sentientia LMS - Moodle 5.2 Ninja/RDS Deployment Guidebook
Audience	IT / Infrastructure team (temporary ninja-server rehearsal)
Package	`Sentientia-LMS-5.2-Complete-Standalone-2026-08-05.zip`
Package size	154 MB (161,989,771 bytes)
SHA-256	`a3697d7e028364a991c5fb7288d1da3756aeadda476382794e693234559284f3`
Target platform	Moodle 5.2 (branch 502) - PHP 8.3 + MySQL 8.4 (AWS RDS)
Prepared	2026-08-05 (L&D Engineering) - Owner: Nitin Rajput
Status	Deploy candidate - static-validated (0 blockers) + all fixes through 2026-08-05 (UI-NAV wave + ADR-028 AI gateway/G.4 publisher, AI features mock-mode flag-OFF); ninja run = first 5.2 runtime validation

> **Live deploy is gated.**
> This guidebook covers a **temporary ninja / sandbox** deployment only. Production go-live on airpay.academy remains a separate, owner-gated decision. Nothing here should be pointed at production data or DNS without explicit sign-off.

Contents

1. Overview & deployment topology

This package is the **complete Moodle 5.2 web root with all Sentientia LMS customisations already applied** - the platform core, all bundled PHP dependencies (vendor libraries under `lib/`), the `theme_sentientia` theme, 46 `local_sentientia_*` plugins, 6 blocks, the Airpay payment gateway, the proctoring quiz-access rule, and the certificate tool. No GitHub checkout, no Composer install, and no Node build are required on the server - the tree is ready to run.

Three things are deliberately **not** in the package and must be provided per-instance (covered below): the site `config.php` (contains DB credentials), the `moodledata` directory, and the database itself.

Target topology

Tier	Component	Notes
Edge	ALB / Nginx (TLS 443)	Terminates HTTPS; forwards to the app server. Certbot on the box is fine for a ninja.
App	EC2 - Apache 2.4 + PHP 8.3	Serves the docroot `.../moodle5.2/public`. Holds the code + moodledata.
Data	AWS RDS - MySQL 8.4	Managed database. Reached privately over port 3306 from the app security group only.

Fresh install vs. data restore
This guide follows the **fresh-install** path (empty database, new admin account) as requested. If you instead need to rehearse the live-data migration, restore the provided DB + moodledata backup *before* running the upgrade step - see §5.6 note.

2. Prerequisites (hard requirements)

Moodle 5.2 will not run on the current production stack.
It requires **PHP 8.3+** and **MySQL 8.4+ / MariaDB 10.11+**. PHP 8.2 or MySQL 8.0 will fail the installer's environment check and the site will not boot. Provision the ninja to the versions in the table below - do not match the current prod RDS (8.0.44) or local (PHP 8.2).

Component	Required	Notes
Operating system	Ubuntu 24.04 LTS (or 22.04) / Amazon Linux 2023	64-bit. Examples below use Ubuntu/apt; translate to dnf on AL2023.
PHP	8.3.x (8.3.6+)	CLI + web SAPI must match. 8.4 also acceptable if all extensions load.
Database	AWS RDS MySQL 8.4 (or MariaDB 10.11+)	utf8mb4 throughout. See §3.2-3.3.
Web server	Apache 2.4 (mod_php or php-fpm) or Nginx 1.24+ + php-fpm	mod_rewrite / try_files required for clean URLs.
Memory	4 GB RAM min (8 GB recommended)	PHP memory_limit 256M+; opcache 256M.
Disk	25 GB+ on the app server	Code ~1.5 GB extracted + moodledata (grows with uploads).
Outbound HTTPS	Allowed (443)	For Moodle updates check, OAuth, and optional AI/WhatsApp features (all default OFF).

3. AWS infrastructure - EC2 app server + RDS database

3.1 EC2 (application server)

Setting	Ninja value
Instance type	t3.large (2 vCPU / 8 GB) recommended; t3.medium minimum
AMI	Ubuntu Server 24.04 LTS (x86_64)
Root volume	gp3, 30 GB
Security group (app)	Inbound: 443 + 80 from ALB/world; 22 from admin IP only. Outbound: all.

3.2 RDS (MySQL 8.4)

Setting	Ninja value
Engine	MySQL 8.4.x (NOT 8.0). MariaDB 10.11+ is an alternative engine.
Instance class	db.t3.medium (2 vCPU / 4 GB) is ample for a ninja
Storage	gp3, 50 GB, autoscaling optional
Multi-AZ	No (single-AZ is fine for a temporary sandbox)
Public access	No - private subnet; reachable only from the app SG
DB name	Leave blank at create time; create it explicitly (§3.5) with utf8mb4
Master user	e.g. `sentientia_admin` (used once to create the app DB + user)
Backups	Default; for a throwaway ninja you may set retention low

3.3 RDS parameter group (critical for Moodle)

Create a custom **DB parameter group** (family `mysql8.4`) and set the values below, then attach it to the instance and reboot. Defaults are close, but character set and packet size must be correct *before* the schema is created.

Parameter	Value	Why
character_set_server	utf8mb4	Moodle 5.x mandates full utf8mb4 (4-byte) storage.
collation_server	utf8mb4_unicode_ci	Consistent collation for all tables.
innodb_file_per_table	1	Required for large-row DYNAMIC format tables.
max_allowed_packet	67108864 (64M)	Large content / backup rows. 8.4 defaults to 64M - confirm.
innodb_buffer_pool_size	~70% of instance RAM	Primary performance lever.
wait_timeout	28800	Avoid premature disconnects during long CLI upgrades.
log_bin_trust_function_creators	1	Only if a restore that recreates routines errors; harmless to set.

3.4 Security groups

- **RDS SG**: inbound TCP 3306 *from the app-server SG id only* (never 0.0.0.0/0).
- **App SG**: inbound 443/80 from the load balancer (or world for a quick ninja), 22 from your admin IP.

3.5 Create the application database + user

Connect to the RDS endpoint with the master user and run:

```
mysql -h <RDS-ENDPOINT> -u sententia_admin -p

CREATE DATABASE sententia
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

CREATE USER 'sententia_app'@'%' IDENTIFIED BY '<STRONG-PASSWORD>';
GRANT ALL PRIVILEGES ON sententia.* TO 'sententia_app'@'%';
FLUSH PRIVILEGES;
```

> **Credential handling**
> Generate the DB password in your secrets manager. It goes only into `config.php` on the app server (file mode 0640, owner www-data). Never commit it, never place it in the repo, and never paste it into a browser field.

4. App-server preparation

4.1 Install PHP 8.3 + extensions (Ubuntu)

```
sudo add-apt-repository ppa:ondrej/php   # if 8.3 not in base repo
sudo apt update
sudo apt install -y apache2 \
  php8.3 libapache2-mod-php8.3 php8.3-cli \
  php8.3-mysql php8.3-xml php8.3-mbstring php8.3-curl \
  php8.3-zip php8.3-gd php8.3-intl php8.3-soap \
  php8.3-bcmath php8.3-ldap php8.3-opcache
sudo a2enmod rewrite ssl headers expires
php -v      # confirm 8.3.x
```

The full required/recommended extension list is in **Appendix B**. The web installer (§6) re-checks every one and reports any that are missing.

4.2 PHP settings (php.ini for both CLI and Apache SAPI)

Directive	Value	Notes
memory_limit	256M (512M rec.)	Some reports/backups need headroom.
max_input_vars	5000	Large admin forms exceed the 1000 default.
post_max_size	100M	Course / SCORM uploads.
upload_max_filesize	100M	Match or below post_max_size.
max_execution_time	300	CLI is unlimited; this is for web.
date.timezone	Asia/Kolkata	Set explicitly to avoid warnings.
opcache.enable	1	Strongly recommended in production mode.
opcache.memory_consumption	256	Large codebase.
opcache.max_accelerated_files	20000	Moodle ships thousands of files.
opcache.revalidate_freq	60	Set 2-5 for a ninja you will edit; 60 for stable.

4.3 Apache virtual host (docroot = the public/ folder)

Docroot must be the public/ sub-folder.
Moodle 5.x uses the public-folder split. Point the web root at `.../moodle5.2/public`, NOT at `.../moodle5.2`. Serving the parent exposes config.php and CLI scripts.

```
<VirtualHost *:443>
  ServerName ninja.sententia.example
  DocumentRoot /var/www/sententia/moodle5.2/public

  <Directory /var/www/sententia/moodle5.2/public>
    Options FollowSymLinks
    AllowOverride All
    Require all granted
  </Directory>

  # Clean URLs (Moodle "slasharguments")
  # AllowOverride All lets Moodle's .htaccess handle rewriting.

  SSLEngine on
  SSLCertificateFile      /etc/letsencrypt/live/ninja.../fullchain.pem
  SSLCertificateKeyFile   /etc/letsencrypt/live/ninja.../privkey.pem
</VirtualHost>
```

4.4 TLS

- Quick ninja: `sudo certbot --apache -d ninja.sententia.example`.
- AWS-native: terminate TLS at an ALB with an ACM certificate and forward 80/443 to the instance.
- Moodle must know it is on HTTPS - set `$CFG->wwwroot` to the `https://` URL (§5.4).

5. Installation - step by step (fresh install)

5.1 Upload & extract

```
sudo mkdir -p /var/www/sententia
# copy the zip to the server, then:
cd /var/www/sententia
unzip /tmp/Sententia-LMS-5.2-Complete-Standalone-2026-08-05.zip
# yields:  /var/www/sententia/moodle5.2/  (with public/ inside)
```

5.2 Create moodledata OUTSIDE the docroot

```
sudo mkdir -p /var/sententiadata
sudo chown -R www-data:www-data /var/sententiadata
sudo chmod 0770 /var/sententiadata
```

5.3 Create config.php from the template

Copy `public/config-dist.php` to `moodle5.2/config.php` and set the key values:

```
$CFG->dbtype    = 'mysqli';
$CFG->dblibrary = 'native';
$CFG->dbhost    = '<RDS-ENDPOINT>';
$CFG->dbname    = 'sententia';
$CFG->dbuser    = 'sententia_app';
$CFG->dbpass    = '<STRONG-PASSWORD>';
$CFG->prefix    = 'mdl_';
$CFG->dboptions = ['dbcollation' => 'utf8mb4_unicode_ci'];

$CFG->wwwroot   = 'https://ninja.sententia.example';
$CFG->dataroot  = '/var/sententiadata';
$CFG->admin     = 'admin';
// require_once(__DIR__ . '/public/lib/setup.php');  // keep the template's tail
```

```
sudo chown -R www-data:www-data /var/www/sententia
sudo chmod 0640 /var/www/sententia/moodle5.2/config.php
```

5.4 Run the fresh install (CLI)

From the project root, create the schema + the admin account in one shot:

```
cd /var/www/sententia/moodle5.2
sudo -u www-data php admin/cli/install_database.php \
  --lang=en --adminuser=admin \
  --adminpass='<ADMIN-PASSWORD>' \
  --adminemail='it@airpay.co.in' \
  --fullname='Sententia LMS (Ninja)' \
  --shortname='Sententia' \
  --agree-license
```

Data-migration alternative (§1).
For a data-restore rehearsal instead of a fresh install: import the provided DB dump into RDS and copy the moodledata (including `filedir/`) into `/var/sententiadata` BEFORE this step, then run `php admin/cli/upgrade.php --non-interactive` in place of install_database.php.

5.5 Purge caches & (optional) post-deploy CLIs

```
sudo -u www-data php admin/cli/purge_caches.php

# optional, safe to run; needed after data restore, harmless on fresh:
sudo -u www-data php public/local/sententia_platform/cli/repair_task_registrations.php --apply
sudo -u www-data php public/local/sententia_catalog/cli/enable_oneclick_enrol.php
```

5.6 Cron (mandatory)

Moodle needs cron every minute or scheduled tasks, notifications, and completion stop working:

```
sudo crontab -u www-data -e
# add:
* * * * * /usr/bin/php /var/www/sententia/moodle5.2/admin/cli/cron.php >/dev/null 2>&1
```

6. Post-install verification

Log in as `admin` and confirm, in order:

#	Check	Expected
1	Site administration → Notifications	No environment failures; "Database" + "PHP version" rows green.
2	Site administration → Server → Environment	PHP 8.3 OK, MySQL 8.4 OK, all required extensions OK.

#	Check	Expected
3	Dashboard loads for admin	Sentientia theme renders; no raw {{ }}; browser console 0 errors.
4	Open a course + a plugin page	Pages render; AMD modules boot (charts / cart badge active).
5	Run the link/persona gate	From `moodle-enhancement/tools/gap-test/` - overwhelmingly healthy.
6	Feature flags	Gap-cohort features default OFF (§7) - confirm before any demo.

This is the first real 5.2 runtime.
The candidate passed static validation (php -l, schema, deprecation, AMD parity - 0 blockers) but has not run on 5.2 before. Treat the ninja boot as validation: capture any Notifications-page warning and the Apache error log, and report before pointing anyone at it.

7. Feature flags & the competitive gap cohort

The package bundles an 11-feature "competitive gap" cohort (AI authoring, skills intelligence, content marketplace, predictive analytics, agentic assistant, xAPI/LRS, talent mobility, public API/LTI, etc.). Every one ships **feature-flagged OFF and in mock mode** (no external AI spend). Default behaviour matches today's Airpay Academy.

- Keep all gap flags OFF for the initial ninja boot - verify the core platform first.
- Flags live in `local_sentientia_core` feature_flags (per-customer + per-tenant overridable).
- AI / WhatsApp / ElevenLabs integrations need real API keys in a server `.env` and stay dormant until both the key and the flag are set.

8. Security & hardening

- **config.php** mode 0640, owner www-data. Never world-readable.
- **moodledata** outside the docroot (§5.2). Confirm `https://.../sentientiadata` is NOT web-reachable.
- **RDS** reachable only from the app SG; no public access; TLS to the DB if your policy requires it.
- **HTTPS only** - redirect 80→443; set an HSTS header once you are confident.
- **Admin account** - strong password; enable 2FA (tool_mfa is bundled) before any external exposure.
- **Debugging OFF** - do not enable developer debug on an internet-facing ninja.
- **Secrets** - DB password + any API keys via the OS/secure store, never in the repo tree.

9. Teardown (temporary ninja)

When the rehearsal is complete:

- Terminate the EC2 instance (or stop it to preserve the volume for a later run).
- Delete the RDS instance - for a throwaway ninja you may skip the final snapshot.
- Remove `/var/sentientiadata` if it held any real learner data.
- Revoke the ninja DNS record and the TLS certificate if issued per-host.

10. Troubleshooting

Symptom	Likely cause	Fix
Installer aborts on "PHP version"	PHP < 8.3	Install/enable PHP 8.3; confirm CLI and web SAPI both report 8.3.
Installer aborts on database version	RDS is MySQL 8.0	Recreate RDS on the mysql8.4 engine (§3.2).
"Incorrect string value" on install	DB not utf8mb4	Fix the parameter group (§3.3), recreate the DB with utf8mb4, reinstall.
config.php exposed / CLI in browser	Docroot points at parent	Set docroot to .../moodle5.2/public (§4.3).
Pages load but no charts / dead buttons	AMD not served or caches stale	Confirm files exist under amd/build/; run purge_caches.php; hard-refresh.
Clean URLs 404	mod_rewrite off / AllowOverride None	a2enmod rewrite; AllowOverride All; reload Apache.
Notifications / completion not firing	Cron not running	Add the www-data crontab entry (§5.6); check cron.php runs.
"Cannot write to dataroot"	moodledata perms	chown www-data + chmod 0770 on /var/sentientiadata.

Appendix A - RDS parameter-group cheat sheet

Parameter	Value
character_set_server	utf8mb4
collation_server	utf8mb4_unicode_ci
innodb_file_per_table	1
max_allowed_packet	67108864
innodb_buffer_pool_size	~70% of RAM (bytes)
wait_timeout	28800
log_bin_trust_function_creators	1 (if restore needs it)

Appendix B - PHP extension checklist

Required (installer blocks without these): `ctype, curl, dom, gd, iconv, intl, json, mbstring, openssl, pcre, SimpleXML, soap, spl, xml, xmlreader, zip, mysqli`.

Recommended: `opcache, ldap, exif, fileinfo, sodium, tokenizer`.

Verify: `php -m | sort`, or read the Notifications / Environment page after install (it flags any gap explicitly).

Appendix C - Package manifest & contents

Attribute	Value
File	Sentientia-LMS-5.2-Complete-Standalone-2026-08-05.zip
Size	154 MB (161,989,771 bytes)
SHA-256	a3697d7e 028364a9 91c5fb72 88d1da37 56aeadda 47638279 4e693234 559284f3
Extracts to	moodle5.2/ (public/ inside = docroot)
Includes	Moodle 5.2 core + all bundled vendor libs; theme_sentientia; 46 local_sentientia_* plugins; 6 sentientia_* blocks; paygw_airpay; sentientia_proctoring; tool_certificate
Excludes	config.php (secret); moodledata / filedir; database; node_modules + dev tooling; stale/quarantine dirs
Verified	php -l 1,567/0 errors; schema well-formed; 0 theme_airpayux survivors; no DB-cred config, no .env/keys in zip

Verify the download before extracting:

```
# Linux
sha256sum Sentientia-LMS-5.2-Complete-Standalone-2026-08-05.zip
# must equal:
# a3697d7e028364a991c5fb7288d1da3756aeadda476382794e693234559284f3
```

Appendix D - Command quick-reference

```
# 1. extract
cd /var/www/sentientia && unzip <package>.zip
# 2. moodledata
sudo mkdir -p /var/sentientiadata && sudo chown www-data:www-data /var/sentientiadata && sudo chmod 0770 /va
# 3. config.php  (edit dbhost/dbname/dbuser/dbpass/wwwroot/dataroot)
sudo cp moodle5.2/public/config-dist.php moodle5.2/config.php
# 4. ownership
sudo chown -R www-data:www-data /var/www/sentientia && sudo chmod 0640 moodle5.2/config.php
# 5. fresh install
cd moodle5.2 && sudo -u www-data php admin/cli/install_database.php --adminpass='***' --adminemail='it@airpa
# 6. caches + cron
sudo -u www-data php admin/cli/purge_caches.php
sudo crontab -u www-data -e   # * * * * * php /var/www/sentientia/moodle5.2/admin/cli/cron.php >/dev/null 2>
```

End of guidebook. Package + this PDF are in the deploy bundle. Live production go-live remains owner-gated.